Category Theory for Mathematical Physics — Exercise Sheet 3

Monoidal categories and string diagrams

Variant: Hints and answers

Conventions as in Lecture 3: pictures read upward, the first factor on the left; Vec_G^ω and its cocycle condition as in Definition 3.7 and Theorem 3.8. The group $\mathbb{Z}_N$ is written additively, with elements $0, 1, \dots, N-1$ and $[x]_N$ the remainder of an integer x mod N; $\zeta_N = e^{2\pi i/N}$. In Exercise 4, as in Lecture 3, $\mathbb{Z}_2 = \{1, s\}$ is written multiplicatively, and the *semion category* is Vec_G^ω on it with $\omega(s,s,s) = -1$. Here ε is the carry of Exercise 3 (not the toric-code anyon); e and u are the units of two operations in Exercise 2, and u is a unit of $\mathbb{Z}_N$ in Exercise 6; in Exercise 5, φ is a group automorphism (not the golden ratio).

Exercise 1 (Interchange and pictures). Let $f\colon X \to X'$, $g\colon X' \to X''$, $h\colon Y \to Y'$ and $k\colon Y' \to Y''$.

(i) Draw $(g \otimes k) \circ (f \otimes h)$ and $(g \otimes \mathrm{id}_{Y''}) \circ (\mathrm{id}_{X'} \otimes k) \circ (f \otimes h)$, and show that both equal $(g \circ f) \otimes (k \circ h)$.

(ii) In $\mathrm{Mat}_{\mathbb{C}}$ let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Compute $A \otimes B$ and $B \otimes A$ (Kronecker products), and check that $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$.

(iii) Boxes on different strands slide past each other (Equation (3.3)). Can two boxes on the same strand? Answer with A and B.

Hint. Each picture has two strands, X on the left; read it row by row and column by column. In (ii), $A \otimes B$ is the block matrix $(a_{ij}B)$.

Answer. (i) All three are the picture with f then g on the left strand and h then k on the right. (ii) $A \otimes B = \begin{pmatrix} 0 & B \\ B & 0 \end{pmatrix}$, $B \otimes A = \begin{pmatrix} A & 0 \\ 0 & -A \end{pmatrix}$, and they differ. (iii) No: $AB = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = BA$.

Exercise 2 (Eckmann–Hilton, in full). Let M be a set with two binary operations $\cdot$ and $*$, with units u for $\cdot$ and e for $*$, such that

$$(a * b) \cdot (c * d) = (a \cdot c) * (b \cdot d) \quad \text{for all } a, b, c, d \in M.$$

(i) Show that $u = e$.

(ii) Show that $a \cdot b = a * b$ and $a \cdot b = b \cdot a$ for all a, b.

(iii) Apply this to $\mathrm{End}(1)$ in a strict monoidal category, with $\cdot = \circ$ and $* = \otimes$, to recover Theorem 3.6, and draw the computation of (ii) for $s, t \in \mathrm{End}(1)$ as pictures of boxes with no strands. Where does strictness enter?

Hint. (i) Write $u = u \cdot u$ and replace each factor by a $*$-product with e. (ii) Write $a = a * e$ and $b = e * b$, then try $a = e * a$ and $b = b * e$.

Answer. (i) $u = e$. (ii) The two operations coincide and are commutative. (iii) Strictness gives $1 \otimes 1 = 1$, so $\otimes$ is an operation on $\mathrm{End}(1)$, and $s \otimes \mathrm{id}_1 = s = \mathrm{id}_1 \otimes s$.

Exercise 3 (The cocycles ω_p of $\mathbb{Z}_N$). For $b, c \in \{0, \dots, N-1\}$ let $\varepsilon(b,c) = (b + c - [b+c]_N)/N$, the *carry* of $b + c$, and let $\omega_p(a,b,c) = e^{2\pi i\, p\, a(b+c-[b+c]_N)/N^2}$.

(i) Show that $\varepsilon(b,c) \in \{0,1\}$ and $\omega_p(a,b,c) = \zeta_N^{\,p\,a\,\varepsilon(b,c)}$. Hence ω_p depends only on p mod N.

(ii) Show that $\varepsilon(b,c) + \varepsilon(a,[b+c]_N) = \varepsilon(a,b) + \varepsilon([a+b]_N,c)$.

(iii) Draw the pentagon of Definition 3.1 for simple objects g,h,k,m of Vec_G^ω, with each vertex labelled by its bracketing (each is the object δ_{ghkm}) and each arrow by its scalar, and the triangle for $g,1,h$ likewise; read off the conditions of Theorem 3.8. Then show that ω_p is normalised and satisfies the cocycle condition, which for $\mathbb{Z}_N$ written additively reads

$$\begin{aligned}&\omega(a+b,c,d)\,\omega(a,b,c+d)\\ &\quad=\omega(a,b,c)\,\omega(a,b+c,d)\,\omega(b,c,d).\end{aligned}$$

(iv) For $N=3$ and $p=1$, list the triples (a,b,c) with $\omega_1(a,b,c)\neq 1$, and the values there.

Hint. (ii) Both sides count the carries in adding $a+b+c$ in two orders. (iii) Write each factor as ζ_N to a power and use (ii) with (b,c,d); the remaining difference is a multiple of N in the exponent of $\zeta_N^{\,p}$.

Answer. (i) $b+c=[b+c]_N+N\varepsilon(b,c)$ with $0\le b+c\le 2N-2$, so $\varepsilon\in\{0,1\}$. (ii) Both sides equal $(a+b+c-[a+b+c]_N)/N$. (iii) The pentagon gives the cocycle condition, the triangle $\omega(g,1,h)=1$; for ω_p the two sides' exponents differ by $p\,N\varepsilon(a,b)\varepsilon(c,d)$, a multiple of N. (iv) The six triples with $a\in\{1,2\}$ and $(b,c)\in\{(1,2),(2,1),(2,2)\}$, where $\omega_1=\zeta_3^{\,a}$.

Exercise 4 (The forgetful functor as a monoidal functor). (i) Show that the forgetful functor $U\colon \mathrm{Rep}(G)\to \mathrm{Vect}_{\mathrm{fd}}$, with J and J_0 identities, is monoidal (Definition 3.10).

(ii) Let $F\colon \mathrm{Vec}_G^\omega\to\mathrm{Vect}_{\mathrm{fd}}$ send a graded space to its underlying space, so $F(\delta_g)=\mathbb{C}$, and give it $J_{g,h}=j(g,h)\,\mathrm{id}_{\mathbb{C}}$ (with $\mathbb{C}\otimes\mathbb{C}=\mathbb{C}$) and $J_0=j_0\,\mathrm{id}_{\mathbb{C}}$, for nonzero numbers $j(g,h)$, j_0. Show that (F,J,J_0) is monoidal exactly when $\omega=\delta j$, with δj as in Lecture 3, and $j(1,g)=j(g,1)=j_0^{-1}$.

(iii) Show that for $G=\mathbb{Z}_2$ and $\omega(s,s,s)=-1$ no such (F,J,J_0) exists.

Hint. (i) $\mathrm{Rep}(G)$'s associator and unitors are those of Vect. (ii) On simple objects every arrow is a scalar times $\mathrm{id}_{\mathbb{C}}$, and Vect's associator on $\mathbb{C}\otimes\mathbb{C}\otimes\mathbb{C}=\mathbb{C}$ is the identity. (iii) Evaluate δj at (s,s,s).

Answer. (i) $U(\alpha)=\alpha$ and $U(l)=l$, $U(r)=r$, so each condition reads $x=x$. (ii) The first condition reads $\omega(g,h,k)\,j(gh,k)\,j(g,h)=j(h,k)\,j(g,hk)$; the unit conditions read $j(1,g)\,j_0=1=j(g,1)\,j_0$. (iii) $(\delta j)(s,s,s)=j(s,1)/j(1,s)=1\neq -1$.

Exercise 5 (Monoidal equivalences of Vec_G^ω). Let ω and ω' be normalised 3-cocycles on a finite group G. You may use that an equivalence is full, faithful and essentially surjective (Leinster Prop. 1.3.18, the converse of Proposition 1.17).

(i) Let φ be an automorphism of G and $j\colon G\times G\to\mathbb{C}^\times$ satisfy

$$\begin{aligned}&\omega'(\varphi(g),\varphi(h),\varphi(k))\\ &\quad=\omega(g,h,k)\,\frac{j(gh,k)\,j(g,h)}{j(h,k)\,j(g,hk)}\end{aligned}\tag{$*$}$$

for all g,h,k. Show that $F(\delta_g)=\delta_{\varphi(g)}$, extended to direct sums, with $J_{g,h}=j(g,h)$ and $J_0=j(1,1)^{-1}$, is a monoidal equivalence $\mathrm{Vec}_G^\omega\to\mathrm{Vec}_G^{\omega'}$.

(ii) Conversely, let (F,J,J_0) be a monoidal equivalence $\mathrm{Vec}_G^\omega\to\mathrm{Vec}_G^{\omega'}$. Show that there are an automorphism φ and a function j satisfying $(*)$.

(iii) Conclude the classification stated in Lecture 3: Vec_G^ω and $\mathrm{Vec}_G^{\omega'}$ are monoidally equivalent exactly when ω' is obtained from ω by an automorphism of G and a coboundary.

Hint. (i) On simple objects every condition of Definition 3.10 is a scalar equation, as in Exercise 4(ii); for the unit conditions use $(*)$ at $(g,1,h)$. (ii) A full and faithful functor keeps $\mathrm{End}(X)$, which by Theorem 2.13 is $\mathbb{C}$ exactly for simple X; choose isomorphisms $F(\delta_g)\cong\delta_{\varphi(g)}$, and look at what $J_{g,h}$ connects.

Answer. (i) $(*)$ is the first condition of Definition 3.10 on simple objects; the unit conditions hold since $j(g,1)=j(1,1)=j(1,g)$ by $(*)$ at $(g,1,h)$. (ii) F permutes the simple objects by φ; $J_{g,h}\neq 0$ forces $\varphi(gh)=\varphi(g)\varphi(h)$, and the first condition becomes $(*)$. (iii) $(*)$ says $\omega'\circ(\varphi\times\varphi\times\varphi)=\omega\,(\delta j)^{-1}$.

Exercise 6 ⋆. *The classes of* $\mathrm{Vec}^{\omega}_{\mathbb{Z}_N}$ *up to equivalence.* For a normalised 3-cocycle ω on $\mathbb{Z}_N$ and $a\in\mathbb{Z}_N$ let

$$Q_\omega(a)=\prod_{r=0}^{N-1}\omega(a,ra,a),$$

with ra reduced mod N. Two cocycles are *cohomologous* if $\omega'=\omega\,(\delta j)^{-1}$ for some $j\colon\mathbb{Z}_N^2\to\mathbb{C}^\times$ (Lecture 3).

(i) Show that $Q_{\omega'}=Q_\omega$ when ω' and ω are cohomologous.

(ii) Show that $Q_{\omega_p}(1)=\zeta_N^{\,p}$. Conclude that $\omega_0,\dots,\omega_{N-1}$ are pairwise not cohomologous.

(iii) For a unit u mod N let $(u^*\omega)(a,b,c)=\omega(ua,ub,uc)$, the cocycle moved by the automorphism $a\mapsto ua$. Show that $Q_{u^*\omega}(1)=Q_\omega(u)$ and $Q_{\omega_p}(u)=\zeta_N^{\,pu^2}$, with u taken in $\{1,\dots,N-1\}$.

(iv) You may use that every normalised 3-cocycle on $\mathbb{Z}_N$ is cohomologous to some ω_p (EGNO Ex. 2.6.4). With Exercise 5, show that $\mathrm{Vec}^{\omega_p}_{\mathbb{Z}_N}$ and $\mathrm{Vec}^{\omega_{p'}}_{\mathbb{Z}_N}$ are monoidally equivalent exactly when $p'\equiv pu^2$ mod N for some unit u, and count the classes for $N=2,3,4,5$.

Hint. (i) Evaluate δj at (a,ra,a) and watch the product over r telescope. (ii) Only one r has a carry. (iii) As r runs over $0,\dots,N-1$, so does $[ru]_N$.

Answer. (i) $\prod_r(\delta j)(a,ra,a)=1$, the product telescoping. (ii) Only $r=N-1$ has $\varepsilon(r,1)=1$, so $Q_{\omega_p}(1)=\zeta_N^{\,p}$. (iii) Exactly u values of r have $[ru]_N+u\geq N$, so $Q_{\omega_p}(u)=\zeta_N^{\,pu\cdot u}$. (iv) 2, 3, 4 and 3 classes for $N=2,3,4,5$.